

# BostonCART

2025-04-29

## 1. Setup

```
# Load libraries
library(caret)
library(dplyr)
library(caTools)
library(rpart)
library(rpart.plot)
library(ROCR)
library(randomForest)
library(ggplot2)

# Read in the data
data <- read.csv("boston_cleaned.csv")
```

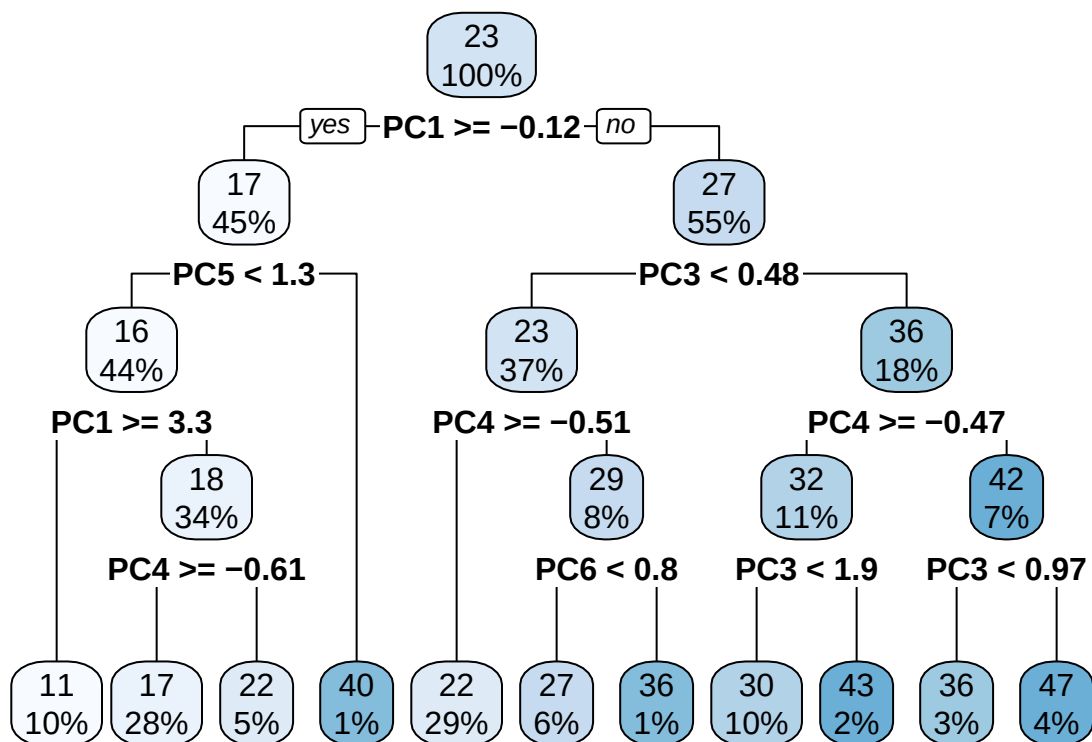
## 2. Beforehand PCA

```
# Run PCA on standardized predictors
pr.out <- prcomp(data %>% select(-MEDV), scale. = TRUE)
pc_data <- as.data.frame(pr.out$x[, 1:6])
pc_data$MEDV <- data$MEDV

# Split the data
set.seed(2641)
spl <- sample.split(pc_data$MEDV, SplitRatio = 0.8)
Train <- subset(pc_data, spl == TRUE)
Test <- subset(pc_data, spl == FALSE)
```

## 3. CART

```
# Build initial CART model
BostonTree <- rpart(MEDV ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6, data = Train,
                    method = "anova", minbucket = 5)
rpart.plot(BostonTree)
```



```

# Make predictions
PredictCART <- predict(BostonTree, newdata = Test)

# Calculate R² for CART model
SSE <- sum((Test$MEDV - PredictCART)^2)
SST <- sum((Test$MEDV - mean(Test$MEDV))^2)
R_squared_CART <- 1 - (SSE / SST)
cat("R-squared for CART model:", R_squared_CART, "\n")

```

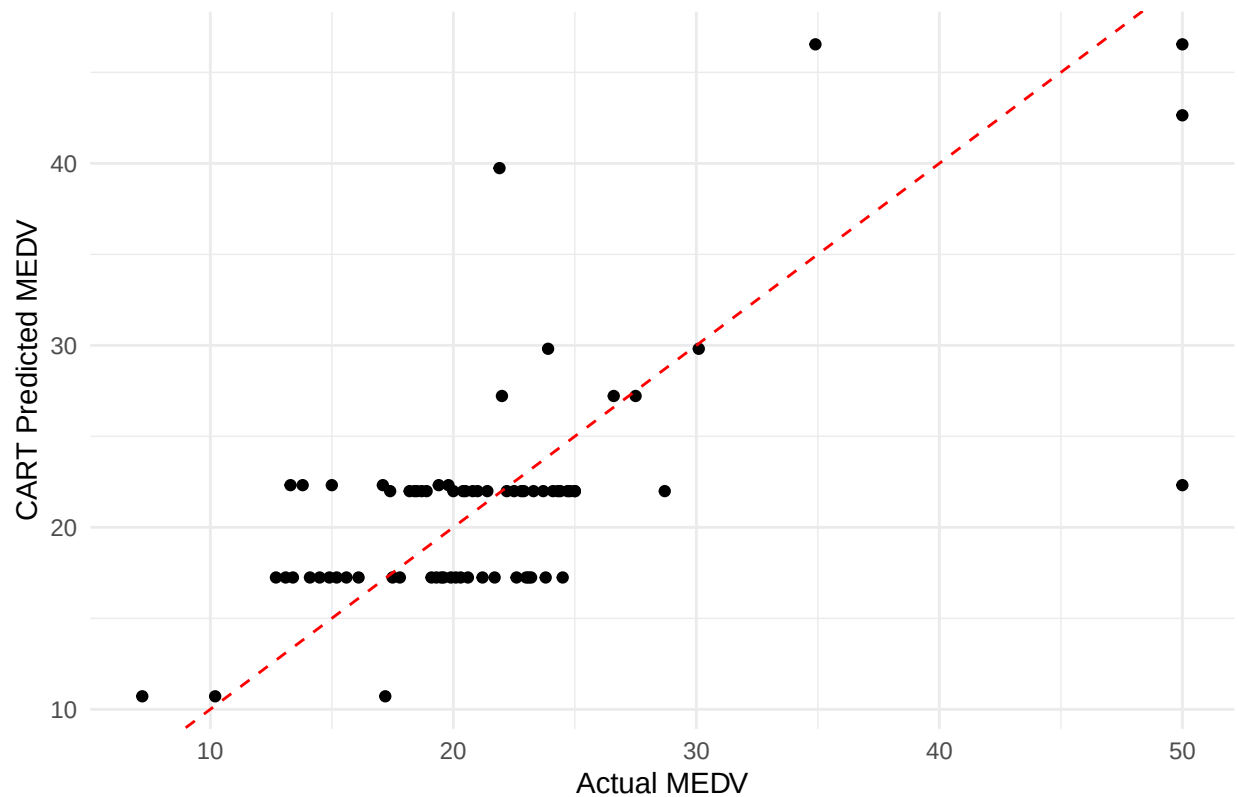
```
## R-squared for CART model: 0.4448521
```

```

#Plotting Actual vs Prediction Value
results <- data.frame(Actual = Test$MEDV, Predicted = PredictCART)
ggplot(results, aes(x = Actual, y = Predicted)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  labs(title = "Actual vs Predicted Values using CART Model",
       x = "Actual MEDV",
       y = "CART Predicted MEDV") +
  theme_minimal()

```

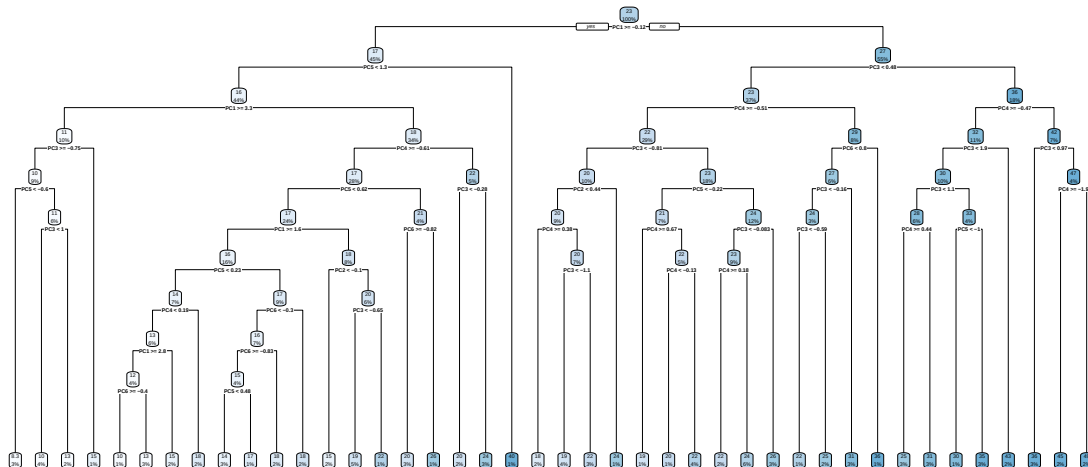
Actual vs Predicted Values using CART Model



#### 4.CROSS-VALIDATION

```
# Cross-validation for the single tree
numFolds <- trainControl(method = "cv", number = 10)
cpGrid <- expand.grid(.cp = seq(0, 0.01, 0.0001))
set.seed(2641)
cv_results <- train(MEDV ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6, data = Train, method = "rpart",
                    trControl = numFolds, tuneGrid = cpGrid)
#Best cv results = 5e-4

# Create a new single CART model based on cross-validation
BostonTreeCV <- rpart(MEDV ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6, data = Train,
                      method = "anova", minbucket = 5, cp=0.0005)
rpart.plot(BostonTreeCV)
```



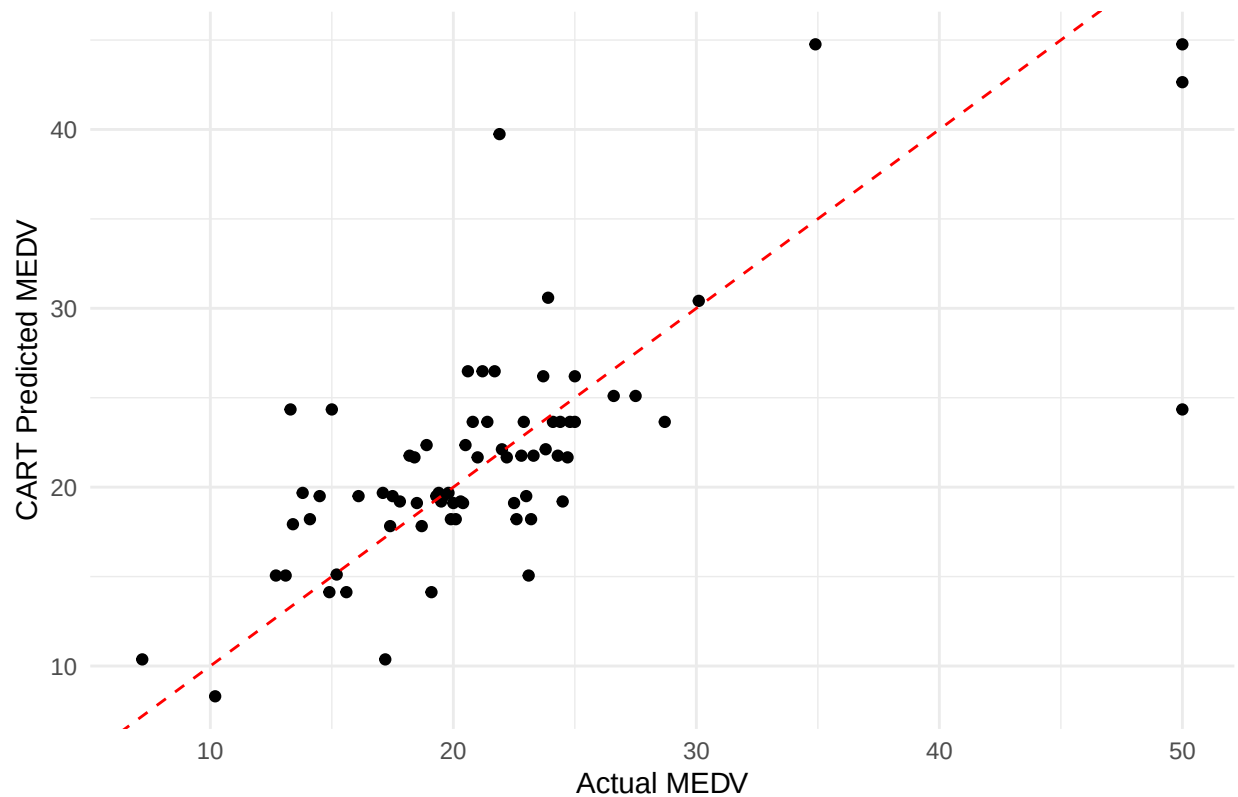
```
# Make predictions
PredictCV <- predict(BostonTreeCV, newdata = Test, type = "vector")

# Calculate SSE, SST, and R2 for CART model with cross-validation
SSE_CV <- sum((Test$MEDV - PredictCV)^2)
SST_CV <- sum((Test$MEDV - mean(Test$MEDV))^2)
R_squared_CV <- 1 - (SSE_CV / SST_CV)
cat("R-squared for CART model with CV:", R_squared_CV, "\n")
```

```
## R-squared for CART model with CV: 0.5044274
```

```
#Plotting Actual vs Prediction Value
resultscv <- data.frame(Actual = Test$MEDV, Predicted = PredictCV)
ggplot(resultscv, aes(x = Actual, y = Predicted)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  labs(title = "Actual vs Predicted Values using cross-validated Model",
       x = "Actual MEDV",
       y = "CART Predicted MEDV") +
  theme_minimal()
```

Actual vs Predicted Values using cross-validated Model



##### 5. Random Forest

```
# Build RF model
BostonForest <- randomForest(MEDV ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6, data = Train,
                             ntree = 300, nodesize = 5)

# Make predictions with RF
PredictForest <- predict(BostonForest, newdata = Test)

# Calculate SSE, SST, and R² for RF model
SSE_RF <- sum((Test$MEDV - PredictForest)^2)
SST_RF <- sum((Test$MEDV - mean(Test$MEDV))^2)
R_squared_RF <- 1 - (SSE_RF / SST_RF)
cat("R-squared for Random Forest model:", R_squared_RF, "\n")
```

## R-squared for Random Forest model: 0.5237854

```
# Create the plot for actual vs. predicted values
results_RF <- data.frame(Actual = Test$MEDV, Predicted = PredictForest)
ggplot(results_RF, aes(x = Actual, y = Predicted)) +
  geom_point(color = "blue") + # Points for actual vs predicted
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") + # Ideal line
  labs(title = "Actual vs Predicted Values using Random Forest Model",
       x = "Actual MEDV",
       y = "Random Forest Predicted MEDV") +
  theme_minimal()
```

Actual vs Predicted Values using Random Forest Model

